

they prefer a VPN service operating in a country like Sweden which has good privacy laws.

TOR

Just using TOR isn't all that hackers do, you gotta use TOR correctly. And it's only good as long as the you ensure traffic to the endpoint router is encrypted as well. And the feds know this which is why a good percentage of the TOR exit routers are monitored. TOR works by anonymising traffic with packet delays and origin obfuscation but lab experiments like the one performed by Sambuddho Chakravarty at the University of Columbia found that by monitoring traffic flow within the network, 100% of the traffic could be identified and the same experiment performed within a real-world TOR network allowed for 81.4 per cent of the traffic to be identified.

INTRUSION DETECTION SYSTEM

High-risk behaviour on the internet always attracts attention, infact, the mere act of downloading the TOR browser or switching on a popular VPN service immediately gets your connection flagged down for monitoring. Hackers need to be one step ahead of the curve so they rely on IDS (Intrusion Detection System) and IPS (Intrusion Prevention System) to figure out if someone is trying to snoop on them and take precautionary measures.

COMMERCIAL PERSONAL SATELLITES

While few have reached this level where they can exploit the inherent design flaws of older commercial satellites to gain control. This is often how command and control server of large botnets are controlled remotely, since the satellite covers a large physical area, it is very difficult to pinpoint where the attacker is hiding. Also, this allows attackers to intercept incoming traffic and redirect them to spoofed pages.

RANDOMISING MAC

One of the most common ways of identifying a networked machine is by looking at the MAC (Media Access Con-

DEFCON



If you wish to keep yourself informed of all the latest developments in the world of black hat hackers then there are a few IRC channels on TOR but that's a close knit community you're never going to get into. Which is why there's DEFCON for the commonfolk. It is one of the largest hacker conventions held annually in Las Vegas and there are plenty of agents from the top security agencies in attendance.

trol) address. Each device is supposed to have its own unique MAC address and a lot of ISPs rely on the same since dynamic IP addressing scheme ensures a different IP address each time a client connects to the network. So what do attackers do? The spoof their network adapter's MAC address as frequently as possible. Every new internet session they join with a device has a new MAC address so that traffic correlation becomes difficult. There are simple batch scripts to do this but some prefer tools like TMAC to do the same.

VIRTUAL MACHINE

Not everyone is well-off to afford a separate machine for their black hat activities. In such cases, attackers rely on Virtual Machines which are preloaded with an arsenal of tools, most of which we've already mentioned. And virtual machines can be destroyed in a manner of seconds without leaving a trace of existence.

DRIVE BYS

Password@123 – if this is your Wi-Fi password then you're among the majority unlucky ones who're an attackers' wet dream. Drive-bys are

when attackers compromise your network and perform an attack and then leave. Your weak Wi-Fi security probably lost a bank a couple of thousand dollars. Needless to say, all machines connected to that network are not part of a botnet. Performing drive-bys are not difficult, there are devices like the Alfa AWUS036H which is a high-gain wireless adapter that can be used from a long range and is compatible with an assortment of malicious tools. One authentication handshake is all that's needed for the attacker to compromise your network.

A GOOD OS

Linux.

DRIVE ENCRYPTION

Let's assume you've goofed up and let your guard down. The authorities are en route with a pair of shiny new ankle bracelets but you wouldn't know till they knock on your front door. So it's best to be prepared and not spoon feed the feds with all the information needed to incriminate you. This is where hard drive encryption matters. Tools like VeraCrypt and AxCrypt allow you to encrypt your hard drive according to military grade spec and then even if caught, it'll take the authorities ages to get in. Anakata, the co-founder of Pirate Bay ended up with a pretty heave jail term because he had not encrypted his drive.

THERE'S A LOT MORE FROM WHERE THAT CAME FROM

Quite a lot of these hackers are extremely proficient at coding and develop their own scripts and tools which we shall never hear about, however, they'll be used to essentially accomplish the same as we have mentioned here. Hackers are not necessarily on the wrong side of the law, it's just that the law is so convoluted and vaguely worded that you're always committing a crime when you're on the fringes and guilt by association is a very common tactic used by the legal system to make hackers turn on their own. So it's only natural for them to take these steps and now you know the how and the why. 